Sales Analysis

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INTRODUCTION

This report presents a comprehensive sales analysis aimed at evaluating overall business performance. It focuses on identifying sales trends, revenue distribution, and key factors influencing sales outcomes. By analyzing historical sales data, the report highlights strengths, weaknesses, and opportunities that can support data-driven decision-making and strategic planning.

OVERVIEW

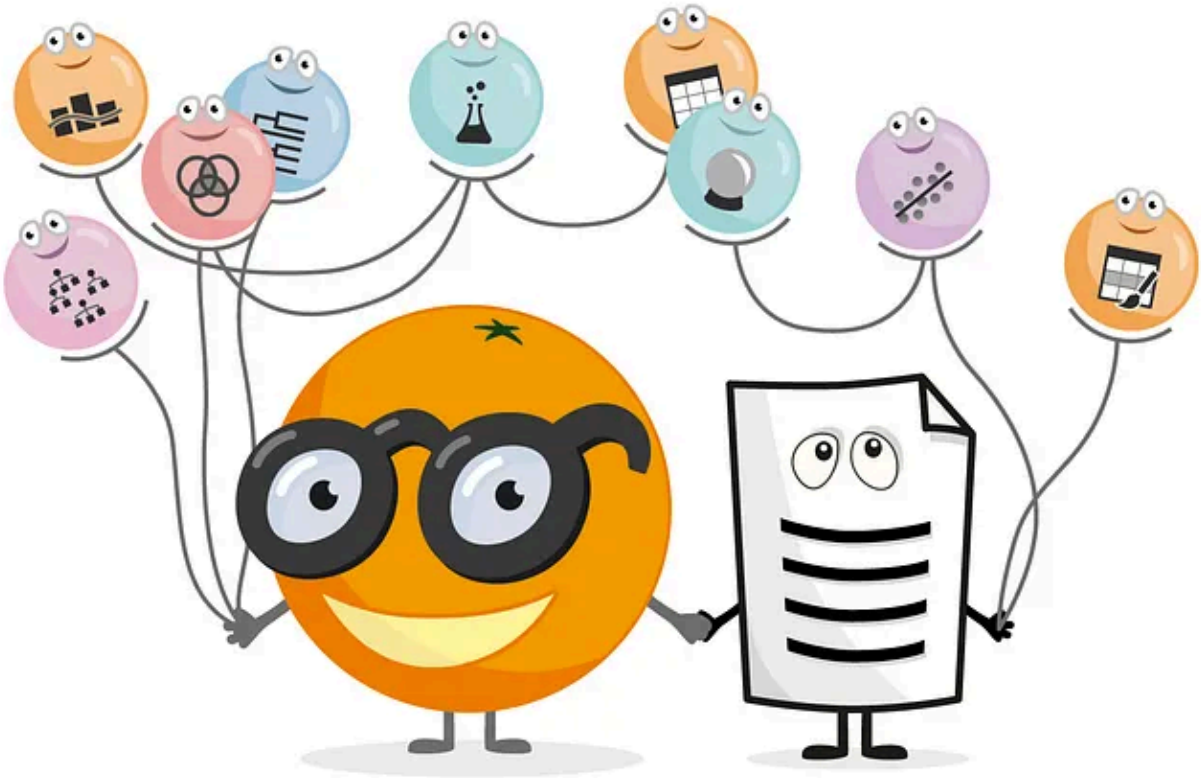
The sales analysis explores key metrics such as total sales, revenue growth, product performance, and regional contributions. Interactive visualizations are used to compare performance across different categories and time periods. The insights derived from this analysis help stakeholders understand customer behavior, optimize sales strategies, and improve overall business efficiency.

DEFINING THE COLUMNS AND DATA TYPES

Overall Column Overview

The dataset contains a comprehensive set of features related to sales operations and business performance. It focuses on key factors that influence sales outcomes, revenue generation, and overall efficiency. The columns include information related to customers, products, sales transactions, pricing, time periods, and regional performance.

By analyzing these columns, the report aims to identify sales trends, performance patterns, and key drivers of revenue. The insights derived support data-driven decision-making, performance evaluation, and the optimization of sales strategies.



DEFINING THE COLUMNS AND DATA TYPES

1. Time & Order Information

- **OrderID:** Unique identifier for each sales transaction.
- **OrderDate:** Date when the order was placed.
- **ShipDate:** Date when the order was shipped.
- **ShipMode:** Shipping method used for the order.

2. Customer Information

- **CustomerID:** Unique identifier for each customer.
- **CustomerName:** Name of the customer.
- **Segment:** Customer segment (e.g., Consumer, Corporate, Home Office).
- **Region:** Geographic sales region.
- **Country:** Country where the customer is located.
- **State:** State or province of the customer.
- **City:** City of the customer.

3. Product Information

- **ProductID:** Unique identifier for each product.
- **ProductName:** Name of the product sold.
- **Category:** High-level product category.
- **SubCategory:** Detailed product classification.

4. Sales & Financial Metrics

- **Sales:** Total revenue generated from the order.
- **Quantity:** Number of units sold.
- **Discount:** Discount applied to the product.
- **Profit:** Net profit earned from the sale.
- **Cost:** Cost associated with the product (if available).

5. Performance & Analysis Fields

- **ProfitMargin:** Percentage of profit relative to sales.
- **OrderPriority:** Priority level assigned to the order.
- **ReturnStatus:** Indicates whether the product was returned (if applicable).

6. Identifiers & Supporting Fields

- **RowID:** Unique row identifier used for internal tracking.
- **Market:** Market or business area where the sale occurred.

BUILDING SCHEMA

A schema is a data modeling technique commonly used in data warehousing and business intelligence. It consists of a **central fact table** that stores quantitative measures, surrounded by multiple **dimension tables** that provide descriptive context.

- **Fact Table:** Contains numerical sales metrics
- **Dimension Tables:** Contain descriptive attributes used for analysis

Sales Data Model Overview

This data model is designed to analyze sales performance across different dimensions such as time, products, customers, and regions. It follows a **schema structure**, enabling efficient querying, simplified relationships, and improved reporting performance in Power BI.

Fact Table

Fact_Sales:

The central table containing key sales measures, including:

- Sales amount
- Quantity sold
- Profit
- Discount

This table serves as the foundation for all sales-related analysis.

Dimension Tables

- **Dim_Date:** Provides time-based attributes such as order date, month, quarter, and year.
- **Dim_Product:** Contains product details including category, subcategory, and product name.
- **Dim_Customer:** Stores customer information such as customer name, segment, and location.
- **Dim_Region:** Represents geographical data including country, region, state, and city.
- **Dim_Shipping:** Includes shipping details such as ship mode and delivery type.

Relationships

The fact table is connected to each dimension table using **foreign keys** (e.g., *DateID*, *ProductID*, *CustomerID*, *RegionID*).
These relationships enable detailed analysis such as:

- Sales by product category
- Profit by region
- Sales trends over time
- Customer segment performance

Purpose of the Model

The purpose of this schema is to support effective sales analysis by identifying trends, measuring performance, and uncovering insights that help optimize sales strategies and business decision-making.

Custom Attributes and Performance Optimization

The data model incorporates several custom attributes to enhance sales analysis and improve overall performance:

▪ Custom Sales Performance Categories

To better analyze revenue trends, sales values were grouped into custom categories such as **Low Sales**, **Medium Sales**, **High Sales**, and **Very High Sales** based on predefined thresholds. This allows for clearer comparison of performance levels across products, regions, and time periods.

▪ Custom Time-Based Grouping

The order date was enhanced with additional time categories, including **Month**, **Quarter**, and **Year**, enabling trend analysis and seasonality insights. These groupings support a deeper understanding of sales behavior over time.

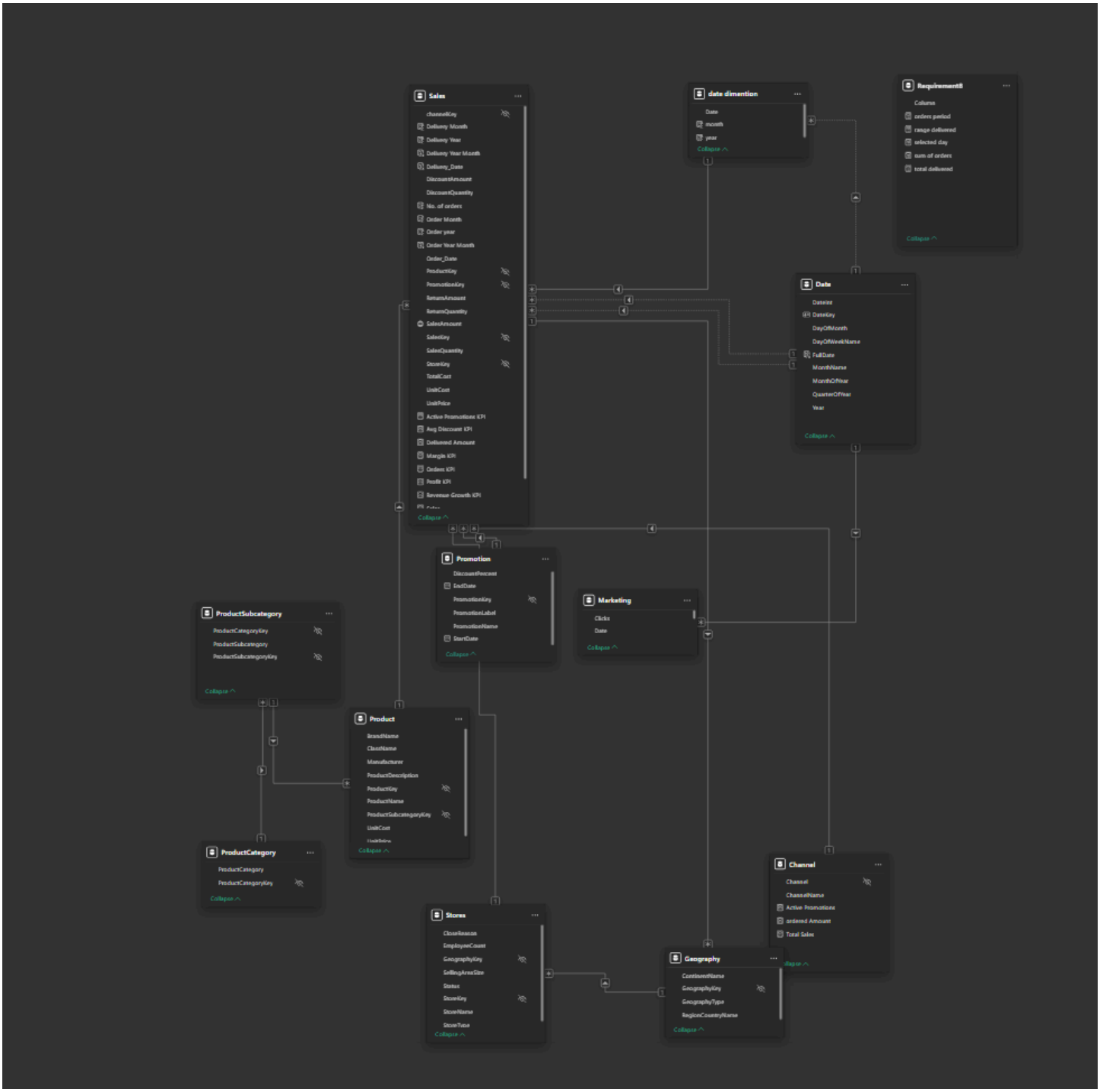
▪ Product and Customer Segmentation

Custom groupings were created for product categories and customer segments to allow more granular analysis of sales distribution. This helps

identify high-performing products, key customer segments, and areas requiring improvement.

▪ Indexing for Faster Joins

To optimize performance, integer-based keys were used to define relationships between the fact table and dimension tables instead of text-based fields. This indexing strategy improves join efficiency, reduces model complexity, and ensures faster query execution in Power BI.



MEASURES USING VARIABLES

Active Promotions: Number of promotions currently running based on start and end dates.

```
Active Promotions =
VAR CurrentDate =
    MAX ( 'Date'[DateKey] )
RETURN
CALCULATE (
    DISTINCTCOUNT ( Promotion[PromotionKey] ),
    FILTER (
        ALL ( Promotion ),
        Promotion[StartDate] <= CurrentDate
            && (
                ISBLANK ( Promotion[EndDate] )
                || Promotion[EndDate] >= CurrentDate
            )
    )
)
```

KPI Achievement %: Percentage of target achieved for the selected KPI.

```
KPI Achievement % =
VAR KPIName =
    SELECTEDVALUE ( KPI_Definition[KPI Name] )

VAR Target =
    SELECTEDVALUE ( KPI_Definition[Target] )

VAR Actual =
    SWITCH (
        KPIName,
        "Total Sales", [Total Sales],
        "Profit", [Profit],
        "Active Promotions", [Active Promotions],
        BLANK()
    )

RETURN
DIVIDE ( Actual, Target )
```

KPI Status: Indicates whether the selected KPI has met its target (1 = achieved, 0 = not achieved).

```
KPI Status =
VAR KPIName =
    SELECTEDVALUE ( KPI_Definition[KPI Name] )

VAR Target =
    SELECTEDVALUE ( KPI_Definition[Target] )

VAR ActualValue =
    SWITCH (
        KPIName,
        "Total Sales",        [Total Sales],
        "Profit",             [Profit],
        "Sales Quantity",     [Sales Quantity KPI],
        "Avg Discount",       [Avg Discount KPI],
        "Orders",             [Orders KPI],
        "Revenue Growth",     [Revenue Growth KPI],
        "Margin",             [Margin KPI],
        "Active Promotions",  [Active Promotions],
        BLANK()
    )

RETURN
IF (
    NOT ISBLANK ( ActualValue )
        && NOT ISBLANK ( Target )
        && ActualValue >= Target,
    1,
    0
)
```

Cost LY YTD YoY %: Percentage change in year-to-date cost compared to the previous year.

```
Cost LY YTD YoY % =
VAR CurrentLYYTD = [Cost LY YTD]
VAR PrevLYYTD =
    CALCULATE(
        [Cost LY YTD],
        FILTER(
            ALL('date dimention'),
            'date dimention'[year] = MAX('date
            dimention'[year]) - 2
        )
    )
RETURN DIVIDE(CurrentLYYTD - PrevLYYTD, PrevLYYTD)
```

Margin LY YTD YoY %: Year-over-year percentage change in year-to-date margin.

```
Margin LY YTD YoY % =
VAR CurrentLYYTD = [Margin LY YTD]
VAR PrevLYYTD =
    CALCULATE(
        [Margin LY YTD],
        FILTER(
            ALL('date dimention'),
            'date dimention'[year] = MAX('date
            dimention'[year]) - 2
        )
    )
RETURN DIVIDE(CurrentLYYTD - PrevLYYTD, PrevLYYTD)
```

Profit LY YTD YoY %: Year-over-year percentage change in year-to-date profit.

```
Profit LY YTD YoY % =
VAR CurrentLYYTD = [Profit LY YTD]
VAR PrevLYYTD =
    CALCULATE(
        [Profit LY YTD],
        FILTER(
            ALL('date dimention'),
            'date dimention'[year] = MAX('date
            dimention'[year]) - 2
        )
    )
RETURN DIVIDE(CurrentLYYTD - PrevLYYTD, PrevLYYTD)
```

Revenue LY YTD YoY %: Year-over-year percentage change in year-to-date revenue.

```
Revenue LY YTD YoY % =
VAR CurrentLYYTD = [Revenue LY YTD]
VAR PrevLYYTD =
    CALCULATE(
        [Revenue LY YTD],
        FILTER(
            ALL('date dimention'),
            'date dimention'[year] = MAX('date
            dimention'[year]) - 2
        )
    )
RETURN DIVIDE(CurrentLYYTD - PrevLYYTD, PrevLYYTD)
```

Clicks YTD % Change: Percentage change in year-to-date clicks compared to last year.

```
Clicks YTD % Change =
VAR Curr = [Clicks YTD]
VAR Prev = [Clicks YTD LY]
RETURN
IF (
    NOT ISBLANK ( Prev ),
    DIVIDE ( Curr - Prev, Prev )
)
```

Orders Period: Total delivered orders for the selected time period.

```
orders period = VAR StartDate = [selected date]
VAR CurrentPeriod = SELECTEDVALUE('time period'[period])
RETURN SWITCH(CurrentPeriod , "week",
    CALCULATE([total delivered],DATESBETWEEN('Date'
    [DateKey], StartDate +1 , StartDate +7)), "15 days",
    CALCULATE([total delivered],DATESBETWEEN('Date'
    [DateKey], StartDate +1 , StartDate +15)) ,
    "month" ,
    CALCULATE([total delivered],DATESBETWEEN('Date'
    [DateKey], StartDate +1 , StartDate +30)) , "6
    months", CALCULATE([total delivered],DATESBETWEEN
    ('Date'[DateKey], StartDate +1 , StartDate +180)))
```

Selected Date: Date constructed from the user's slicer selection.

```
selected date =
VAR SlicerYear = SELECTEDVALUE('Date'[Year])
VAR SlicerMonthOfYear = SELECTEDVALUE('Date'
[MonthOfYear])
VAR SlicerDayOfMonth = SELECTEDVALUE('Date'[DayOfMonth])
RETURN DATE(SlicerYear,SlicerMonthOfYear,
SlicerDayOfMonth)
```

BUILDING DYNAMIC INTERACTIVE DASHBOARD

The Power BI dashboard is designed to analyze **sales performance** by leveraging custom DAX measures and interactive features. Key metrics such as

Total Sales, Revenue Growth, Profit, Margin, Orders, and Active Promotions are calculated using DAX and displayed through various visualizations, including bar charts, line charts, and KPI cards.

Interactive elements such as **filters and slicers** allow users to explore sales data across multiple dimensions, including **time periods, regions, product categories, customer segments, and promotional activity**, enabling deeper insights into sales trends and performance drivers. The dashboard is designed for intuitive navigation, with features such as **drill-through for detailed analysis** and **conditional formatting** to highlight critical metrics, supporting informed, data-driven business decisions.